

Smarter technology for all

Servicing the ThinkSystem SR680a V3 with B200

Lenovo

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Preface

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Current release date: July 2025

Current release level: 1.0

The information in this publication is correct as of the date of the latest revision and is subject to change at any time without notice.

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Prerequisites

- [ES42373B – Intel Xeon processor architecture for ThinkSystem V3 servers](#)
- [ES51757B – Introducing ThinkSystem tools](#)
- [ES52374 – ThinkSystem tools for the ThinkSystem V3 platform](#)
- [ES41759C – ThinkSystem problem determination](#)
- [ES51780C – Servicing the ThinkSystem storage controllers](#)

Objectives

After completing the course, you will be able to:

- Describe the ThinkSystem SR680a V3 with B200 and its components
- List the SR680a V3 with B200 specifications
- Describe the SR680a V3 with B200 configurations
- Describe the SR680a V3 with B200 management tools
- Describe the problem determination steps and explain how to troubleshoot issues with the SR680a V3 with B200

Product overview

Product description and front, rear, and inside views

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SR680a V3 with B200 product overview

The ThinkSystem SR680a V3 with B200 is an 8U, air-cooled, dual-socket rack server designed for high-performance generative AI computing and HPC workloads. It supports two 5th Generation Intel Xeon Scalable processors, 16 2.5-inch hot-swap NVMe drives, 10 front PCIe Gen 5 slots, and eight NVIDIA B200 GPUs. The server features advanced GPU-to-GPU communication and high-speed PCIe Gen 5 connectivity between the GPUs, processors, and networking components.

There are two SR680a V3 with B200 machine types:
7DM9 (three-year warranty)
7DP9 (five-year warranty)



Specifications

Features	SR680a V3 with B200
Form factor	8U rack server
Processor	Support for two 5 th Generation Intel Xeon Scalable processors with up to 64 cores per processor and a TDP of up to 350 W
Memory	32 DIMM slots with two processors (16 DIMM slots per processor) Each processor has eight memory channels with two DIMMs per channel (DPC) Lenovo TruDDR5 RDIMMs are supported DIMMs operate at up to 5600 MHz at 1 DPC and up to 4400 MHz at 2 DPC
Disk drive bays	Up to 16 2.5-inch hot-swap drive bays supporting PCIe Gen 5 NVMe drives
OS boot drive	Support for two M.2 drives with optional Intel VROC NVMe RAID support for OS boot and data storage functions
PCIe slots	Up to 10 PCIe Gen 5 x16 slots in the front • Eight PCIe Gen 5 x16 FHHL slots in the PCIe switch shuttle with GPU direct support • Two PCIe Gen 5 x16 FHHL slots in two PCIe riser cards with CPU direct connection
PSUs	Three power mode settings • Eight hot-swap redundant 3200 W CFFv5 PSUs • Eight 3200 W CFFv5 PSUs, N+N redundancy with OVS (Oversubscription) • Six 3200 W CFFv5 PSUs, N+1 redundancy
Cooling	• Two front primary fans: 60 mm x 56 mm • Fifteen rear primary fans: 80 mm x 56 mm • Four rear auxiliary fans: 40 mm x 56 mm

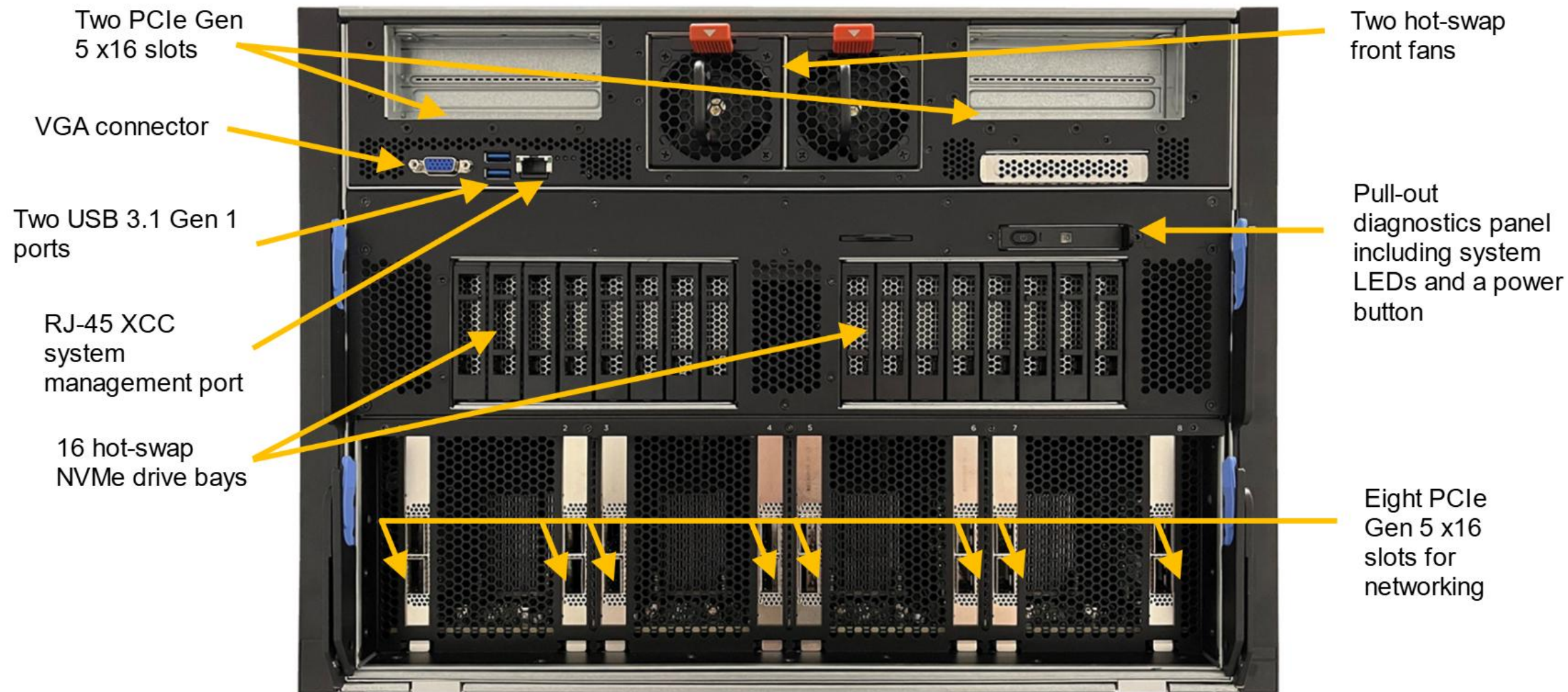


Note: For the latest specifications, refer to the product guide on [Lenovo Press](#).

Specifications

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Cooling	• Two front primary fans: 60 mm x 56 mm • Fifteen rear primary fans: 80 mm x 56 mm • Four rear auxiliary fans: 40 mm x 56 mm Support for N+1 redundancy
GPUs	• Eight NVIDIA B200 1000 W SXM6 GPUs with 180 GB HBM3 memory
Management tools	XClarity series tools: • XClarity Controller 2 (XCC2) • XClarity Provisioning Manager (LXPM) • XClarity Essentials (LXCE)

Front view

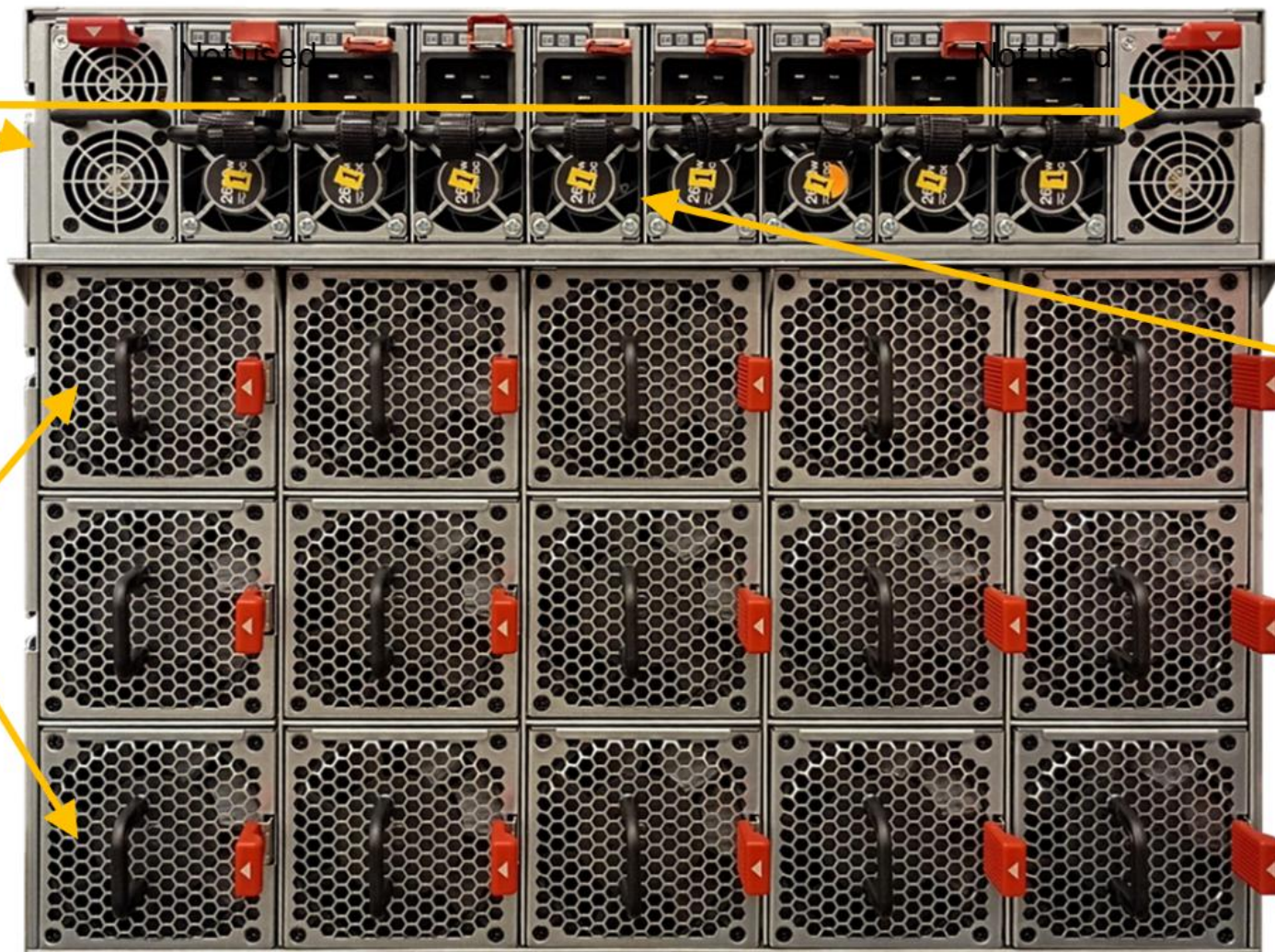


Rear view

Four hot-swap
rear auxiliary
fan

15 hot-swap
rear fans

Eight hot-swap
power
supplies



Inside views

Click the buttons to see inside views of the SR680a V3 with B200.

Exploded view

**B200 GPU
complex assembly**

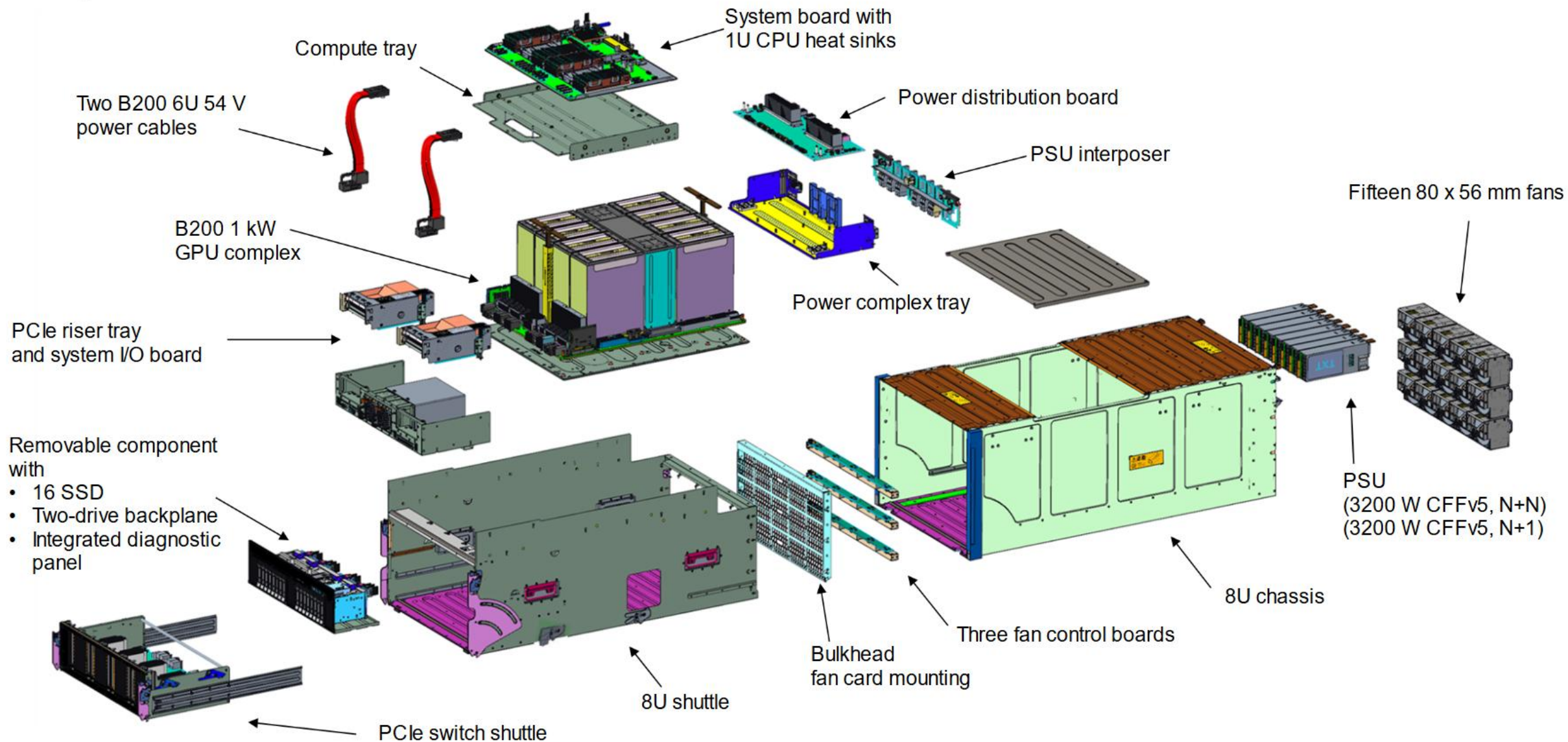
**Power complex
assembly**

**System board
assembly**

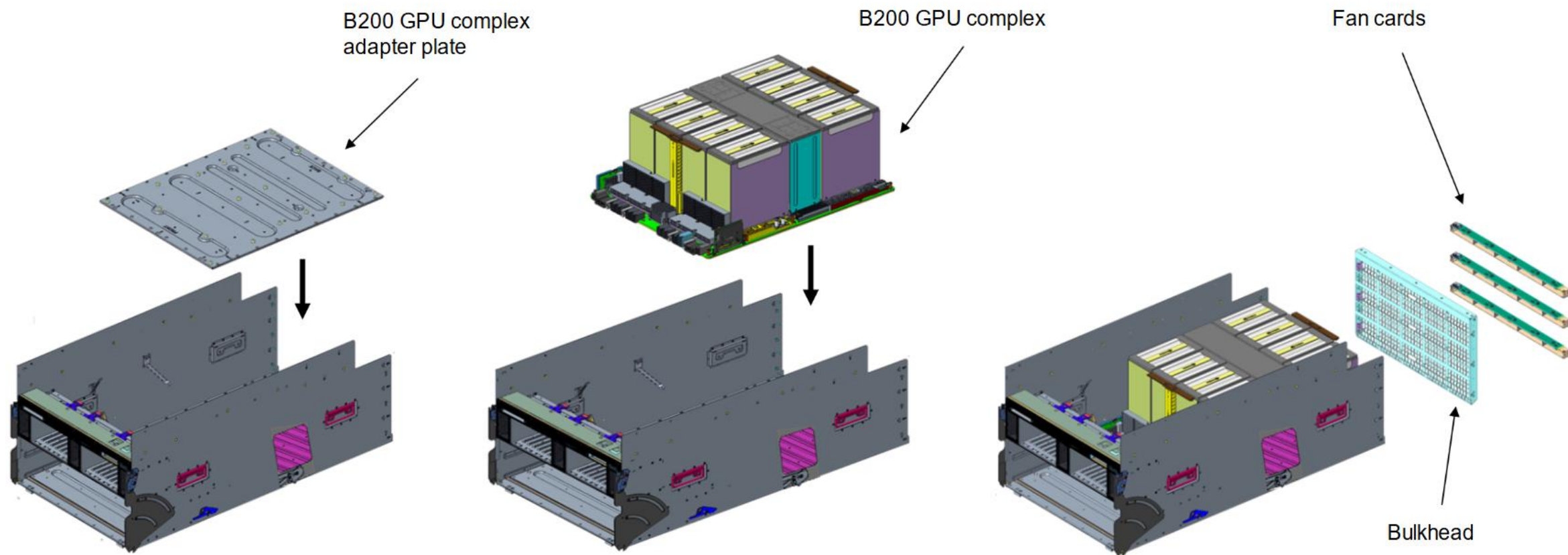
**System shuttle
assembly**



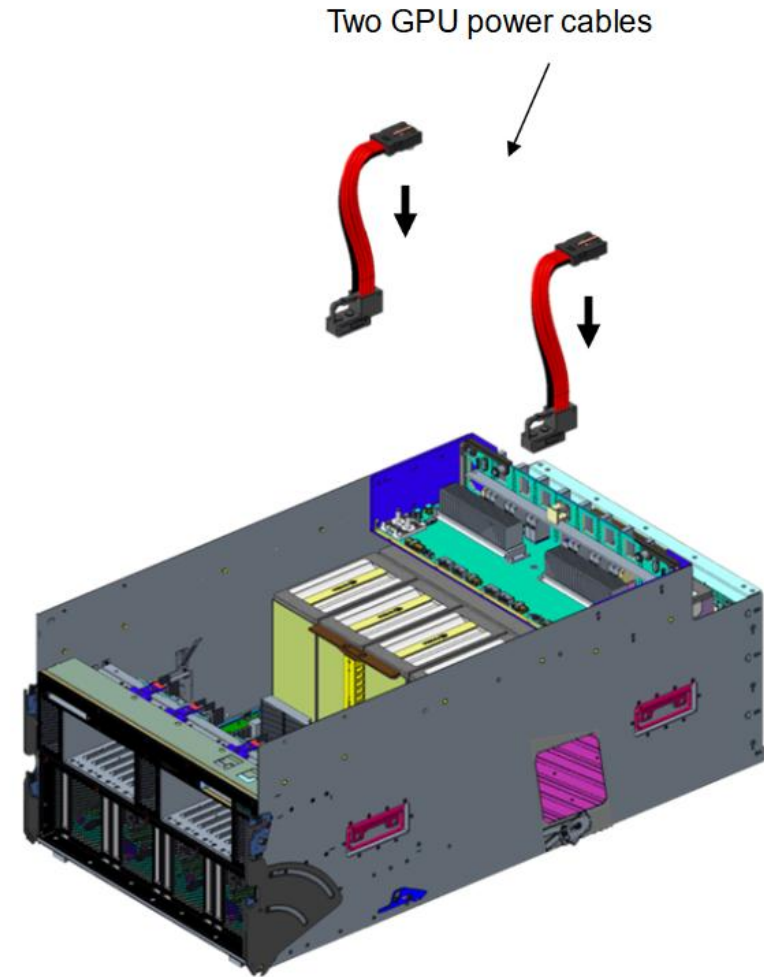
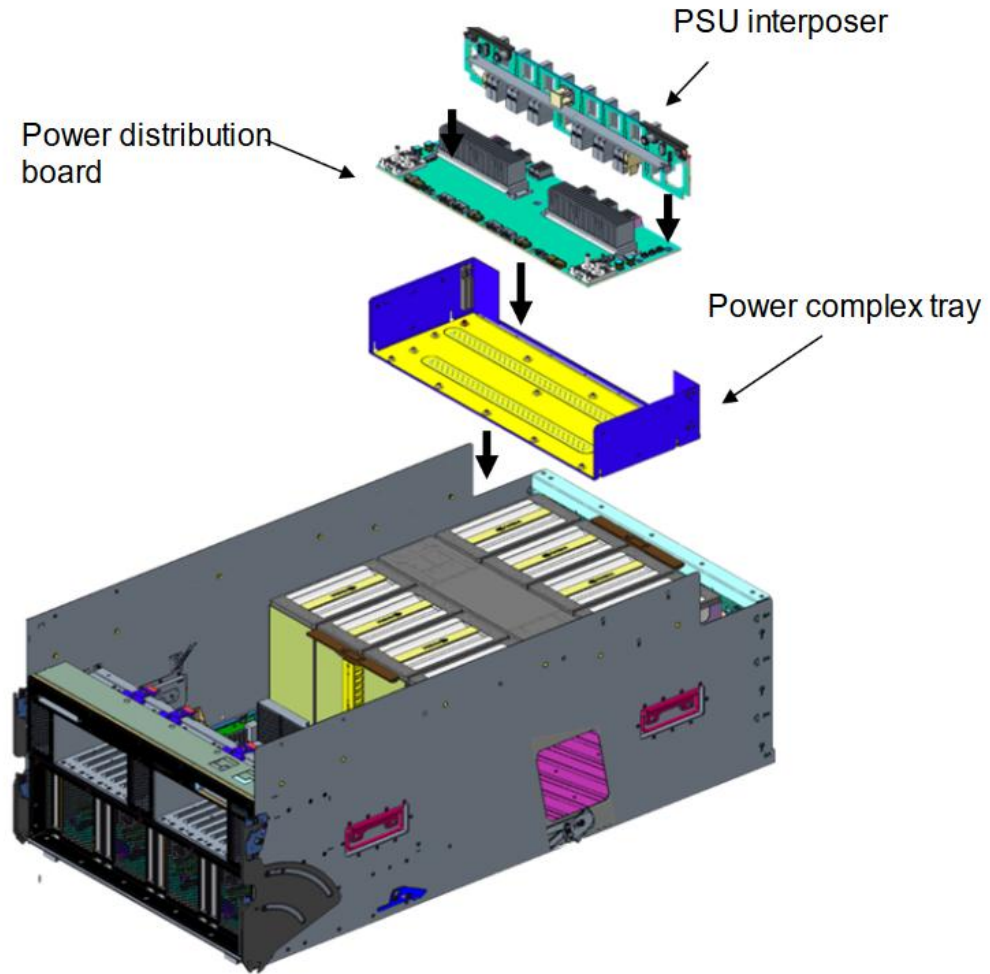
Exploded view



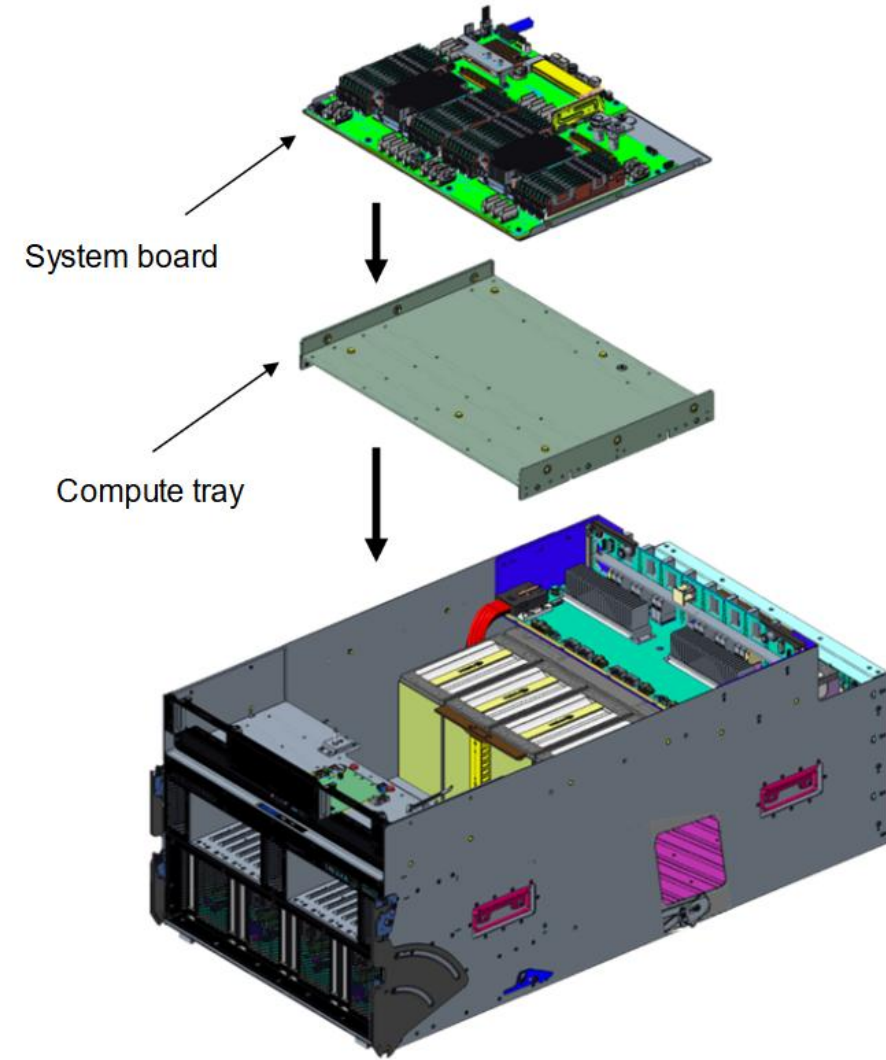
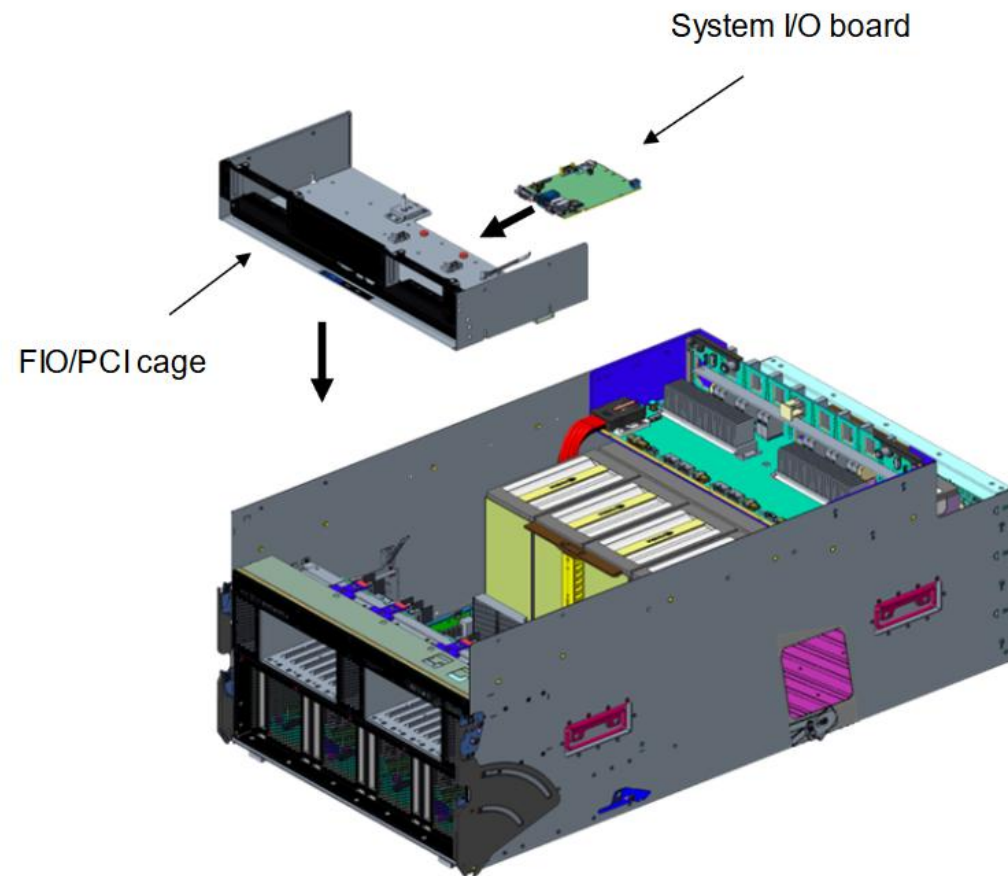
B200 GPU complex assembly



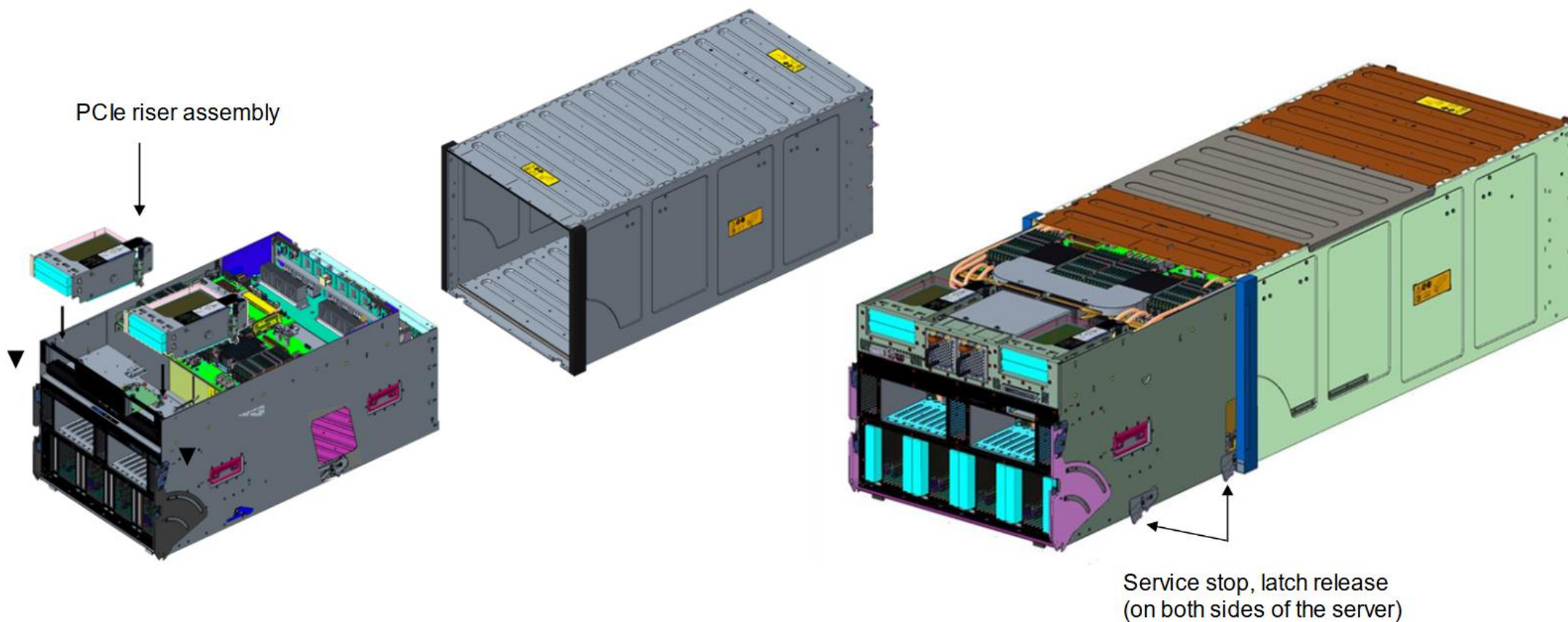
Power complex assembly



System board assembly



System shuttle assembly



Note: PCIe riser assemblies can also be installed before installing a system shuttle.